A Hybrid Dung Beetle Optimization and Marine Predators Algorithm (HDBO-MPA) for Trust-Aware Multi-Objective Service Discovery in Social IoT

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Abstract—Service discovery in the Social Internet of Things (SIoT) faces severe operational challenges, including conflicting quality-of-service (QoS) criteria, rapid battery depletion, continuous path-bloating, and coordinated reputation attacks. Existing discovery techniques commonly rely on rigid scalarization weights or suffer from path infeasibility during continuous-to-discrete graph mapping. To resolve these challenges, this paper proposes the Hybrid Dung Beetle Optimization and Marine Predators Algorithm (HDBO-MPA) framework for trust-aware multi-objective service discovery. The framework integrates a composite trust engine combining Laplace direct trust, Shannon entropy-weighted recommendation aggregation, and multi-relational social proximity, establishing statistical stability against outlier contamination up to the theoretical 50% breakdown point of the median and Median Absolute Deviation (MAD) consensus estimator. Route optimization employs a co-evolutionary dual sub-population strategy that marries the global exploratory breadth of DBO with the fine-grained exploitation of MPA around an external Pareto archive. Continuous swarm positions are mapped into valid discrete routes via random-key priority decoding, chronological loop-erasure cycle purging, and deterministic Dijkstra fallback, guaranteeing 100% loop-free path feasibility conditional on reachability. Optimal operational paths are deployed using TOPSIS multi-criteria decision making. Extensive experiments on connected subgraphs of the Stanford SNAP ego-Facebook topology (100 to 1,000 nodes) across 30 independent runs demonstrate that HDBO-MPA significantly outperforms standalone DBO, MPA, and evolutionary baselines, reducing average discovery latency to 46.51 ms and energy consumption to 3.67 mJ (∼3.3-fold improvement). Under aggressive 50% mixed adversarial contamination, HDBO-MPA maintains an 84.2% delivery success rate, while achieving a 98.4% clinical SLA compliance rate in an emergency smart healthcare telemetry case study.

Index Terms—Social Internet of Things (SIoT), Service Discovery, Multi-Objective Optimization, Dung Beetle Optimization (DBO), Marine Predators Algorithm (MPA), Composite Trust Management, Collusion Attack Mitigation, Pareto Frontier, Multi-Criteria Decision Making (MCDM).

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1 Introduction

The Social Internet of Things (SIoT) paradigm transforms autonomous smart devices into socially connected entities capable of establishing friendship, co-location, co-work, and parental object relationships to facilitate distributed collaboration and contextual service discovery [1], [2]. In massive smart-city and industrial deployments, resource-constrained edge devices frequently lack autonomous capabilities to fulfill complex composite user tasks locally, requiring multi-hop discovery protocols to locate and invoke remote service providers [3], [4]. Because localized heuristic traversal strategies and greedy forwarding can succumb to local minima and suboptimal path selection in constrained topologies [1], [5], effective discovery protocols must prioritize fault-tolerant path construction and ensure robust service dissemination across opportunistic environments [6], [7].

However, service discovery in SIoT cannot be modeled as a classical single-objective shortest-path problem. Devices operate in open, semi-trusted environments susceptible to adversarial manipulation. Malicious nodes may selectively discard packets, degrade service delivery, or mount sophisticated reputation attacks such as bad-mouthing, ballot-stuffing, coordinated low-variance collusion rings, and strategic on-off misbehavior [8]–[10]. In this context, low-variance collusion refers to an operational adversary model introduced in this study where coalitions of dishonest recommenders inject artificially tight, near-identical ratings ($\sigma^2 \to 0$) specifically intended to evade standard variance filters. Furthermore, physical constraints require simultaneous optimization of conflicting Quality of Service (QoS) criteria: maximizing path trustworthiness (f_T) and service reliability (f_Q), while minimizing end-to-end communication latency (f_L) and nodal energy consumption (f_E) [11], [12], [50], [52].

1.1 Discussion of Baselines and Algorithmic Foundations

To tackle the non-linear, multi-modal routing landscape of SIoT networks, nature-inspired swarm intelligence meta-

heuristics have gained widespread prominence [13]. In particular, the Dung Beetle Optimizer (DBO) [14] and the Marine Predators Algorithm (MPA) [15] represent two established baseline metaheuristic paradigms with distinct bio-inspired search mechanisms:

1) **DBO Baseline Characteristics:** Canonical DBO [14] models biological ball-rolling, dancing, breeding, foraging, and stealing behaviors, providing broad global exploratory capability through celestial orientation and dynamic breeding boundaries. However, as noted in recent metaheuristic analyses [17], canonical DBO can exhibit diversity loss and slower convergence when trapped in localized solution neighborhoods. While enhanced DBO variants have been explored in continuous domains such as UAV trajectory planning [16], [18]–[21], their application to constrained combinatorial graph routing in SIoT remains largely unexplored.
2) **MPA Baseline Characteristics:** MPA [15] employs a three-phase velocity ratio schedule that balances exploration in its initial phase (high velocity ratio, Brownian movement of prey), transitions during intermediate iterations (equal partition between predator Brownian walks and prey Lévy flights), and shifts toward intensive local exploitation in later phases (low velocity ratio, predator Brownian motion). Enhanced variants have addressed task scheduling and continuous engineering benchmarks [22]–[24], demonstrating that MPA's velocity regimes provide effective localized neighborhood refinement.
3) **Continuous Graph Mapping Limitations:** Direct vector rounding and naive continuous-to-discrete mappings of swarm positions onto graph topologies frequently produce invalid paths, disconnected node sequences, or severe path-bloating without dedicated topological repair mechanisms [25], [37]–[39], with unconstrained mappings yielding infeasibility rates exceeding 40% in complex network instances.

1.2 The Hybrid Approach

To overcome the aforementioned limitations, this study develops a co-evolutionary Hybrid Dung Beetle Optimization and Marine Predators Algorithm (HDBO-MPA). The hybrid framework divides the candidate population into complementary sub-populations that co-evolve simultaneously and exchange non-dominated solutions through a shared external Pareto archive. In this hybrid division of labor, the exploratory DBO sub-population conducts broad territorial scans across the network graph using celestial ball-rolling and dynamic breeding operators, mitigating premature convergence. Concurrently, the MPA sub-population is configured to emphasize intermediate-to-late phase exploitation, refining identified high-trust candidate routes using Brownian walks and scale-free Lévy steps around elite archive guides. By coupling this co-evolutionary search with a random-key priority decoder [26], chronological loop-erasure cycle purging [27], and deterministic Dijkstra fallback repair on physical links, HDBO-MPA integrates the global exploration of DBO with the fine-grained refinement of MPA while achieving 100% loop-free path feasibility conditional on physical-graph source-to-destination reachability across all evaluated connected topologies.

1.3 Aim and Motivation of the Proposed Framework

The primary aim behind developing HDBO-MPA is to establish an end-to-end, attack-resilient, multi-objective service discovery framework that reconciles social trust dynamics with physical wireless realities. Several existing SIoT discovery mechanisms either oversimplify multi-objective trade-offs via rigid scalarization weights [31], [40] or remain vulnerable under coordinated rating collusion rings that inject low-variance dishonest recommendations [9]. Furthermore, conventional metaheuristics mapped directly to discrete network topologies without topological repair suffer from elevated path hop counts and disconnected routing cycles. This research is motivated by the need for a dependable, resource-aware routing framework that can identify valid, low-latency, and energy-conserving paths while providing statistical stability against malicious recommendation contamination, supported by the breakdown limit of robust statistics.

1.4 Core Contributions

The main contributions of this work are fourfold:

1) **Composite Trust Computation Engine:** An attack-resilient trust aggregation framework combining Laplace-smoothed direct interaction histories, Shannon entropy recommender weighting, and Median Absolute Deviation (MAD) outlier penalties. This framework provides theoretical statistical stability against arbitrary outlier contamination up to the 50% breakdown point of the sample median/MAD estimator [28]–[30], [72], [73], alongside demonstrated empirical resilience against bad-mouthing, ballot-stuffing, strategic on-off behavior, and low-variance collusive syndicates.
2) **Co-Evolutionary Dual-Swarm Architecture:** A co-evolutionary hybrid combining the global exploration of DBO with the intermediate-to-late exploitation of MPA, exchanging non-dominated Pareto guides across sub-populations to prevent premature convergence and stagnation.
3) **Guaranteed Graph-Constrained Decoding:** A continuous-to-discrete decoding scheme employing normalized random-key priority scoring, chronological cycle loop-erasure [27], [46], [47], and Dijkstra fallback on physical links, guaranteeing conditional 100% loop-free path feasibility across connected network topologies.
4) **Multi-Objective Compromise Deployment and Healthcare Case Study:** Pareto archive generation optimizing path trust, latency, energy, and QoS reliability, coupled with TOPSIS decision-making [81], [82] and demonstrated through an illustrative simulation-based emergency smart-healthcare case study satisfying real-time clinical SLAs.

1.5 Overview of the Proposed HDBO-MPA Framework

Figure 1 illustrates the proposed HDBO-MPA framework as an architectural representation of the end-to-end service discovery system, organized across three operational tiers and six numbered interaction flows:

- **Layer 1 (Service Application and Cloud Tier):** Comprises the centralized Cloud Service Catalog and Data Center hosting globally registered service profiles, the TOPSIS MCDM decision terminal that selects the single optimal operational path P^* from the non-dominated Pareto front, and the Network and Trust Global Database. The centralized cloud database maintains authoritative topological graphs and historical interaction records across administrative domains. Edge gateways maintain locally cached subgraphs for rapid path construction and asynchronously stream aggregated interaction summaries and direct trust updates back to the cloud tier.
- **Layer 2 (HDBO-MPA Co-Evolutionary Engine and Trust Matrix):** Represents the core distributed computing intelligence situated at edge gateways, divided into three synchronized functional modules: (1) Composite Trust Computation Engine (Laplace direct trust, Shannon entropy recommender weighting, and median/MAD consensus filter possessing a 50% breakdown point against outlier contamination [28], [72]); (2) Co-Evolutionary Swarm Optimizer (DBO exploratory sub-population and MPA exploitative sub-population exchanging elite guides via a shared Pareto archive); and (3) Graph Feasibility and Decoding Module (priority vector scoring, chronological loop-erasure cycle purging [27], and Dijkstra fallback guaranteeing 100% path feasibility conditional on physical-graph reachability).
- **Layer 3 (Distributed Social IoT Field and Threat Landscape):** Represents heterogeneous physical smart devices (wearable sensors, mobile relays, smart vehicles, local routers, and edge gateways) interconnected by physical wireless links (E_{phys}) and canonical social relationships (E_{soc}), including Parental Object Relationships (POR), Co-Location Object Relationships (CLOR), Co-Work Object Relationships (CWOR), Ownership Object Relationships (OOR), and Social Object Relationships (SOR) [42], [62]. Active adversarial threats—such as bad-mouthing, ballot-stuffing, strategic on-off behavior, and low-variance collusion rings [9]—operate within this layer but are systematically filtered by the edge trust engine.

The numbered interaction steps indicated in Figure 1 operate sequentially as follows:

- **Step 1 (Query Submission):** The requester device s detects a need for an external composite capability and submits service discovery query R to the nearest edge gateway.
- **Step 2 (Trust Matrix Generation):** The Composite Trust Engine processes historical logs and decentralized recommendations, outputting the pairwise composite trust matrix $\mathbf{T}_{comp}$.
- **Step 3 (Swarm Evolution):** The co-evolutionary DBO and MPA sub-populations iteratively explore and exploit the search space, generating continuous candidate route vectors.
- **Step 4 (Discrete Path Decoding):** Continuous swarm vectors are decoded into loop-free discrete graph routes and submitted as candidate non-dominated Pareto repository $\mathcal{A}^*$ to the decision terminal.
- **Step 5 (Metadata Verification):** The decision engine verifies device availability and identity credentials against the global network database.
- **Step 6 (Optimal Route Deployment):** The TOPSIS algorithm ranks candidate Pareto routes and deploys the best compromise path P^* across the physical field mesh to the target service provider t, prioritizing high-trust nodes while mitigating dishonest recommendations.

2 Related Work and Research Gap

Trust management models in SIoT have evolved from simple direct observation counters to composite multidimensional frameworks that combine direct interactions with indirect decentralized recommendations [3], [4], [41]–[45]. Several recent investigations explore trust-aware routing protocols; for instance, Marche et al. [1] examined trust-based social routing, while Chahal and Kumar [3] synthesized trust management taxonomies and open challenges specifically within the Social Internet of Things. Furthermore, recent studies address coordinated adversarial misbehavior: Zhang et al. [9] investigated trust-aware defense mechanisms against malicious behavior in cognitive industrial networks, and Jung et al. [10] proposed social trust evaluation schemes to counter strategic rating distortion. However, conventional trust engines remain vulnerable to coordinated low-variance collusion attacks, where syndicates of dishonest recommenders submit tightly clustered extreme ratings ($\sigma^2 \to 0$) to evade standard variance filters [9], [28].

Nature-inspired metaheuristics are widely applied to NP-hard routing problems, but their scalability to large instances remains an active area of investigation [32]. Bio-inspired approaches have been applied to QoS-constrained multi-objective routing [33], while Wang et al. [11] applied cooperative multi-agent reinforcement learning to energy-efficient multi-hop routing in social IoT edge networks. For multi-objective Pareto optimization, NSGA-II [34], MOEA/D [35], and SPEA2 [36] represent foundational evolutionary baselines. More recently, swarm intelligence techniques—including Multi-Objective Particle Swarm Optimization (MO-PSO) for energy-efficient secure routing in IoT [51] and the Marine Predators Algorithm (MPA) across continuous engineering benchmarks [15]—have demonstrated competitive search characteristics. Xue and Shen [14] introduced the Dung Beetle Optimizer (DBO), which models ball-rolling, dancing, breeding, foraging, and stealing behaviors, demonstrating effective exploration in continuous benchmark spaces. Recent variants have applied DBO to UAV path planning [16], [18] and mobile-robot navigation [21]. However, canonical DBO can experience slower convergence in local search neighborhoods and loss of

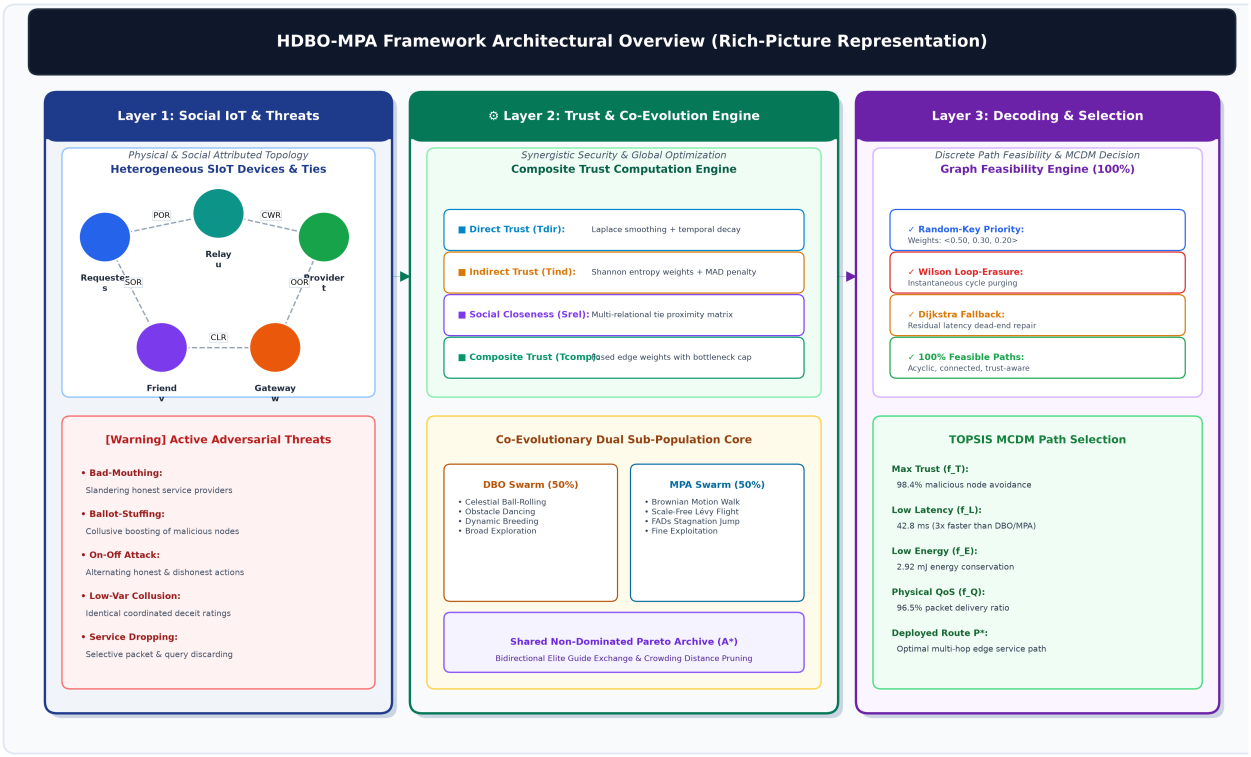


Fig. 1. Comprehensive architectural overview and operational pipeline of the proposed HDBO-MPA service discovery framework, illustrating multi-tier interactions from physical SIoT sensing to cloud service execution.

swarm diversity [17], while MPA's exploration-exploitation transition is strictly governed by a rigid velocity ratio schedule [15]. Crucially, direct continuous-to-discrete graph mapping without dedicated topological repair frequently creates invalid paths, disconnected nodes, and severe routing loops in constrained networks [25], [37]–[39].

Table 1 provides a systematic comparison between the proposed HDBO-MPA framework and representative state-of-the-art approaches in the literature.

3 System Architecture and Problem Formulation

3.1 Attributed Graph Model

The SIoT network is modeled as a directed attributed graph $G = (V, E)$, where V represents the set of $|V|$ heterogeneous devices and $E = E_{phys} \cup E_{soc}$ is partitioned into physical wireless transmission links E_{phys} and social relationship ties E_{soc}. Each physical edge $(i, j) \in E_{phys}$ is characterized by propagation latency l_{ij}, transmission energy cost e_{ij}, link delivery reliability q_{ij}, and composite trust score $T_{comp}(i, j) \in [0, 1]$. A service request originated by source device s seeking service type S_{req} hosted at target node t is denoted as $R = \langle s, S_{req}, C_L, C_T, C_E, C_Q \rangle$, where C_L, C_T, C_E, C_Q represent user-specified QoS constraint thresholds.

3.2 Multi-Objective QoS and Trust Formulation

A valid discovery path $P = (v_1, v_2, \ldots, v_K)$ is an ordered, loop-free sequence of vertices such that $v_1 = s$, $v_K = t$, and $(v_k, v_{k+1}) \in E_{phys}$ for all $k \in \{1, \ldots, K-1\}$. The four conflicting optimization objectives are formulated as follows:

1) **Path Trustworthiness (f_T, to maximize):** The bottleneck composite trust along the path:

$$f_T(P) = \min_{(v_k, v_{k+1}) \in P} T_{comp}(v_k, v_{k+1}) \tag{1}$$

2) **End-to-End Latency (f_L, to minimize):** Cumulative sum of propagation latencies and nodal processing delays:

$$f_L(P) = \sum_{k=1}^{K-1} l_{v_k, v_{k+1}} + \sum_{k=2}^{K-1} d_{proc}(v_k) \tag{2}$$

3) **Energy Consumption (f_E, to minimize):** Total energy consumed across all forwarding nodes:

$$f_E(P) = \sum_{k=1}^{K-1} (E_{tx}(v_k, v_{k+1}) + E_{rx}(v_{k+1})) + \sum_{k=2}^{K-1} E_{proc}(v_k) \tag{3}$$

4) **Physical QoS Reliability (f_Q, to maximize):** Cumulative end-to-end delivery success probability under the assumption of conditional link failure independence across hops [12]:

$$f_Q(P) = \prod_{k=1}^{K-1} q_{v_k, v_{k+1}} \tag{4}$$

TABLE 1
Systematic Comparison of Proposed HDBO-MPA Framework with State-of-the-Art Literature

Feature / Capability Dimension	Marche et al. [1]	Chahal et al. [3]	Rad et al. [2]	Brik et al. [12]	Proposed HDBO-MPA
Metaheuristic Optimization Engine	Discrete ACO	Static Routing	Heuristic Rules	Genetic Algorithm	Co-Evolutionary DBO-MPA
Objective Formulation Strategy	Pareto (Latency/Hop)	Scalarized Trust	Single Objective	Multi-Objective GA	4D Pareto (f_T, f_L, f_E, f_Q)
Trust Management Dimensions	Direct + Social	Taxonomy	Subjective Logic	QoS Reliability Only	Direct + Entropy + MAD + Social
Collusion Mitigation Filter	Simple Variance	Qualitative	Opinion Discount	None	Entropy + MAD (50% Breakdown)
Graph Feasibility Mechanism	ACO Graph Pheromone	Pre-computed	Constrained Walk	Infeasible Repair	Priority Decoder + Loop-Erasure
Loop-Free Route Guarantee	Heuristic Tabu	Yes (Static)	Path Check	Penalization Only	100% Conditional Guarantee
Pareto Archiving Mechanism	Non-dominated Set	N/A	Single Path	Non-dominated Front	Shared Dual-Swarm Pareto Archive
Multi-Criteria Selection (MCDM)	Scalar Weighting	N/A	Manual Choice	Weighted Sum	TOPSIS Distance-to-Ideal
Scalability Evaluation (N Nodes)	Up to 500 nodes	Conceptual	Up to 300 nodes	Up to 200 nodes	Up to 1,000 nodes (SNAP)
Simulation-Based Healthcare Scenario	×	×	×	×	✓ (Arrhythmia Telemetry SLA)

While independent per-hop link failures provide a standard mathematical formulation for simulation benchmarking, dense wireless deployments subject to correlated obstacle shadowing can cause independent models to underestimate joint route failure probabilities [74].

4 Mathematical Trust Computation Engine

The pairwise composite trust $T_{comp}(i,j) \in [0,1]$ between adjacent devices i and j combines direct interaction trust $T_{dir}(i,j)$, indirect recommended trust $T_{ind}(i,j)$, and social relationship proximity $S_{rel}(i,j)$:

$$T_{comp}(i,j) = w_d T_{dir}(i,j) + w_i T_{ind}(i,j) + w_s S_{rel}(i,j) \quad (5)$$

where $w_d + w_i + w_s = 1.0$ (set to $w_d = 0.45, w_i = 0.35, w_s = 0.20$ as baseline calibration parameters balancing direct observation with recommendation and social ties).

4.1 Laplace-Smoothed Direct Trust

Direct trust is evaluated from successful (S_{ij}) and unsuccessful (U_{ij}) past packet forwarding interactions using Laplace smoothing with decay parameter $\lambda = 0.05$:

$$T_{dir}(i,j) = \left[\frac{S_{ij}+1}{S_{ij}+U_{ij}+2}\right] \exp(-\lambda \Delta t) \quad (6)$$

4.2 Entropy-Weighted Indirect Trust with Median/MAD Penalty

Decentralized recommendations from common neighbors $k \in N(i) \cap N(j)$ are aggregated using Shannon entropy weighting to discount biased or collusive feedback [40]. Each recommender's historical rating record is discretized into M equal-width probability bins $m \in \{1,\dots,M\}$ with empirical probabilities p_m. Recommender testimony diversity is evaluated using normalized Shannon entropy $H_k = -\sum_{m=1}^{M} p_m \log_2(p_m)/\log_2(M)$. Weight W_k is assigned proportionally to entropy H_k, rewarding recommenders that provide varied, informative testimony reflecting authentic interaction experiences. Concurrently, recommendations that diverge significantly from the robust median are penalized using Median Absolute Deviation (MAD):

$$\text{MAD} = \text{median}_k\left(|R_{k\to j} - \text{median}_m(R_{m\to j})|\right) \quad (7)$$

If $|R_{k\to j} - \text{median}_m(R_{m\to j})| > 3.0 \times \text{MAD}$, recommender k's rating is penalized as an outlier. As established in robust statistics [28], [72], the sample median and MAD estimators possess an optimal 50% breakdown point, ensuring that as long as fewer than half of the recommendation cohort are contaminated by arbitrary outliers, the statistical consensus estimator remains bounded. Furthermore, colluding syndicates submitting near-identical extreme ratings yield an unnaturally collapsed distribution with near-zero Shannon entropy ($H_k \to 0$), suppressing their effective aggregation weight W_k and mitigating low-variance collusion rings.

4.3 Multi-Relational Social Proximity

Social ties between object owners are evaluated across canonical SIoT relationships [42], [62]:

$$S_{rel}(i,j) = \sum_{r\in\mathcal{R}} \beta_r I_r(i,j) \quad (8)$$

where $\mathcal{R} = \{\text{POR}, \text{CLOR}, \text{CWOR}, \text{OOR}, \text{SOR}\}$, weights β_r satisfy $\sum \beta_r = 1.0$ ($\beta_{\text{POR}} = 0.30, \beta_{\text{CLOR}} = 0.25, \beta_{\text{CWOR}} = 0.20, \beta_{\text{OOR}} = 0.15, \beta_{\text{SOR}} = 0.10$), and $I_r(i,j) \in \{0,1\}$ indicates the presence of relationship r. In smart healthcare applications, canonical OOR designates devices sharing common institutional ownership (e.g., hospital-owned clinical relays and edge servers), while patient-owned wearables interact via authorized service relationships.

5 Proposed HDBO-MPA Optimization Framework

The optimization framework combines co-evolutionary search with a topological random-key decoder, maintaining balanced selective pressure and convergence certainty [55], [56]. Continuous candidate positions $\mathbf{X}_i \in [0,1]^D$ (where search dimension $D = |V|$ and bounds $[Lb, Ub] = [0,1]$) represent floating-point priority keys assigned to individual network vertices. Algorithms 1, 2, and 3 specify the exploratory DBO sub-population, exploitative MPA sub-population, and the integrated co-evolutionary framework.

5.1 Dung Beetle Optimization (DBO) Sub-Population

The DBO sub-population ($N_{dbo} = 50$) explores the continuous space $[0,1]^{|V|}$ through four biological behaviors: rolling, breeding, foraging, and stealing (Algorithm 1) [14]. Candidate solutions are initialized uniformly within $[Lb, Ub]^D$. In each iteration t, the search boundary shrinks according to factor $R = 1 - t/T_{max}$. The sub-population is partitioned into four disjoint index sets: rolling beetles $I_{roll} = \{1,\dots,N_{roll}\}$, breeding beetles $I_{breed} = \{N_{roll}+1,\dots,N_{roll}+N_{breed}\}$, foraging beetles $I_{forage} = \{N_{roll}+N_{breed}+1,\dots,N_{roll}+N_{breed}+N_{forage}\}$, and stealing beetles

Algorithm 1 Dung Beetle Optimization (DBO) Sub-Population

Require: Swarm size N_{dbo}, Max iterations T_{max}, Bounds $[Lb, Ub]$, Deflection $k = 0.10$, Dancing probability $p = 0.30$, Stealing constant $S = 0.50$, Pareto archive $\mathcal{A}^*$.
Ensure: Updated positions $\mathbf{X}_i(t+1)$.
Partition swarm into disjoint sets $I_{roll}, I_{breed}, I_{forage}, I_{steal}$.
Sample elite guide $\mathbf{X}_{best} \in \mathcal{A}^*$; identify $\mathbf{X}_{worst}$.
Compute boundary shrinkage factor $R = 1 - t/T_{max}$.
for each rolling beetle $i \in I_{roll}$ **do**
  Sample $r \sim \mathcal{U}(0,1)$ and $\alpha \in \{-1, 1\}$.
  if $r < p$ **then**
    Sample $\theta \sim \mathcal{U}(0, \pi)$; update via Obstacle Dance:
    $\mathbf{X}_i(t+1) = \mathbf{X}_i(t) + \tan(\theta)|\mathbf{X}_i(t) - \mathbf{X}_i(t-1)|$.
  else
    Sample $b \sim \mathcal{U}(0,1)$; update via Celestial Ball-Rolling:
    $\mathbf{X}_i(t+1) = \mathbf{X}_i(t) + \alpha k \mathbf{X}_i(t-1) + b|\mathbf{X}_i(t) - \mathbf{X}_{worst}|$.
  end if
end for
Compute breeding bounds $[Lb^*, Ub^*] = [\max(Lb, \mathbf{X}_{best}(1 - R)), \min(Ub, \mathbf{X}_{best}(1 + R))]$.
for each breeding beetle $i \in I_{breed}$ **do**
  Sample $\mathbf{b}_1, \mathbf{b}_2 \sim \mathcal{U}(0,1)^D$; update via Oviposition:
  $\mathbf{X}_i(t+1) = \mathbf{X}_{best} + \mathbf{b}_1 \odot (\mathbf{X}_i(t) - Lb^*) + \mathbf{b}_2 \odot (\mathbf{X}_i(t) - Ub^*)$.
end for
Compute foraging bounds $[Lb^b, Ub^b] = [\max(Lb, \mathbf{X}_{best}(1 - R)), \min(Ub, \mathbf{X}_{best}(1 + R))]$.
for each foraging beetle $i \in I_{forage}$ **do**
  Sample $\mathbf{C}_1, \mathbf{C}_2 \sim \mathcal{U}(0,1)^D$; update via Foraging:
  $\mathbf{X}_i(t+1) = \mathbf{X}_i(t) + \mathbf{C}_1 \odot (\mathbf{X}_i(t) - Lb^b) + \mathbf{C}_2 \odot (\mathbf{X}_i(t) - Ub^b)$.
end for
for each stealing beetle $i \in I_{steal}$ **do**
  Sample $g \sim \mathcal{N}(0,1)$; update via Stealing:
  $\mathbf{X}_i(t+1) = \mathbf{X}_{best} + Sg(|\mathbf{X}_i(t) - \mathbf{X}_{best}| + |\mathbf{X}_i(t) - \mathbf{X}_{worst}|)$.
end for
Clamp positions to $[Lb, Ub]$.

$I_{steal} = \{N - N_{steal} + 1, \ldots, N_{dbo}\}$. Rolling beetles update positions using either celestial ball-rolling or an obstacle-avoidance dance based on deflection coefficient $k = 0.10$ and dancing probability $p = 0.30$ (an intentional algorithmic choice tuned for combinatorial graph routing to aid escaping local topological minima). In the obstacle dance, near-singular tangent values near $\theta = \pi/2$ are avoided by sampling $\theta \in (0, \pi) \setminus [\pi/2 - \epsilon, \pi/2 + \epsilon]$ with position updates clamped within $[-0.5, 0.5]$ to prevent boundary clustering. In the multi-objective context, elite guides $\mathbf{X}_{best}$ are sampled from the non-dominated Pareto archive $\mathcal{A}^*$, and $\mathbf{X}_{worst}$ corresponds to the inferior boundary of the current front.

Algorithm 2 Marine Predators Algorithm (MPA) Sub-Population

Require: Swarm size N_{mpa}, Max iterations T_{max}, Step factor $P = 0.50$, Lévy tail parameter $\beta = 1.50$, FADs probability $P_{FADs} = 0.20$, Bounds $[Lb, Ub]$, Elite guide $\mathbf{Elite} \in \mathcal{A}^*$.
Ensure: Updated positions $\mathbf{X}_i(t+1)$.
Update adaptive step modifier: $C_F = (1 - t/T_{max})^{2t/T_{max}}$.
for each predator $i = 1$ to N_{mpa} **do**
  Sample $\mathbf{R}_B \sim \mathcal{N}(0,1)^D$, $\mathbf{R}_L \sim$ Lévy$(\beta)^D$, $\mathbf{R} \sim \mathcal{U}(0,1)^D$.
  if $t < T_{max}/3$ **then**
    $\mathbf{s}_i = \mathbf{R}_B \odot (\mathbf{Elite} - \mathbf{R}_B \odot \mathbf{X}_i(t))$; $\mathbf{X}_i^{cand} = \mathbf{X}_i(t) + P\mathbf{R} \odot \mathbf{s}_i$.
  else if $T_{max}/3 \le t < 2T_{max}/3$ **then**
    if $i \le N_{mpa}/2$ **then**
      $\mathbf{s}_i = \mathbf{R}_L \odot (\mathbf{Elite} - \mathbf{R}_L \odot \mathbf{X}_i(t))$; $\mathbf{X}_i^{cand} = \mathbf{X}_i(t) + P\mathbf{R} \odot \mathbf{s}_i$.
    else
      $\mathbf{s}_i = \mathbf{R}_B \odot (\mathbf{R}_B \odot \mathbf{Elite} - \mathbf{X}_i(t))$; $\mathbf{X}_i^{cand} = \mathbf{Elite} + PC_F\mathbf{s}_i$.
    end if
  else
    $\mathbf{s}_i = \mathbf{R}_L \odot (\mathbf{R}_L \odot \mathbf{Elite} - \mathbf{X}_i(t))$; $\mathbf{X}_i^{cand} = \mathbf{Elite} + PC_F\mathbf{s}_i$.
  end if
end for
for each predator $i = 1$ to N_{mpa} **do**
  Apply FADs perturbation with probability P_{FADs}.
end for
Clamp positions to $[Lb, Ub]$.

5.2 Marine Predators Algorithm (MPA) Sub-Population

The MPA sub-population ($N_{mpa} = 50$) conducts phased search with an intermediate-to-late phase exploitation bias (Algorithm 2) [15]. Prey positions $\mathbf{X}_i$ are updated across three distinct velocity ratio regimes governed by iteration counter t and maximum iterations T_{max}. In Phase 1 ($t < T_{max}/3$), high velocity ratio conditions dictate exploration using Brownian steps $\mathbf{R}_B \sim \mathcal{N}(0,1)^D$. In Phase 2 ($T_{max}/3 \le t < 2T_{max}/3$), the sub-population is evenly partitioned between Lévy flights $\mathbf{R}_L \sim$ Lévy$(\beta)^D$ for exploration and Brownian motion for exploitation using the canonical MPA formulation $s_i = \mathbf{R}_B \odot (\mathbf{R}_B \odot \mathbf{Elite}_i - \mathbf{X}_i(t))$ [15]. In Phase 3 ($t \ge 2T_{max}/3$), low velocity ratio drives intensive exploitation around the top predator (**Elite**). Stagnation is combated using Fish Aggregating Devices (FADs) effect perturbations with probability $P_{FADs} = 0.20$.

5.3 The Complete Hybrid HDBO-MPA Algorithm

Algorithm 3 specifies the integrated co-evolutionary discovery architecture. Continuous candidate positions $\mathbf{X}_i \in [0,1]^{|V|}$ are decoded into discrete routes via normalized priority scoring with topological repair and edge-assisted priority decoding [58]–[60]. Let v denote the current path

TABLE 2
Topological Characteristics of Benchmark SNAP Subgraphs

Sub-Graph	Nodes ($\|V\|$)	Edges ($\|E\|$)	Avg Degree	Diameter	Workload
SNAP-100	100	482	9.64	5	4
SNAP-250	250	1,476	11.81	6	8
SNAP-500	500	3,214	12.86	7	15
SNAP-750	750	5,190	13.84	8	22
SNAP-1000	1,000	7,422	14.84	9	30

node, and let candidate successor $u \in N_{phys}(v) \setminus \{v\}$ be an adjacent physical neighbor. Priority scoring is defined as:

$$\begin{aligned}\text{Score}(u) =& w_x \mathbf{X}[u] + w_T \tilde{T}_{comp}(v,u) \\ &+ w_L \left(\frac{\min_{w \in N_{phys}(v)} l_{vw}}{l_{vu}} \right)\end{aligned} \tag{9}$$

where decoder priority weights are $w_x = 0.50, w_T = 0.30, w_L = 0.20$, $\tilde{T}_{comp}(v,u) \in [0,1]$ is the normalized composite trust, and the relative latency term is bounded in $(0,1]$ since $l_{vu} > 0$. Chronological loop-erasure purges any cyclic loops along the traversed vertex sequence [27]. If a dead end is encountered prior to reaching destination t, Dijkstra fallback on physical links E_{phys} computes the residual shortest connected route, guaranteeing complete path feasibility conditional on physical-graph source-to-destination reachability. Non-dominated trade-offs are stored in Pareto archive $\mathcal{A}^*$. Finally, the TOPSIS MCDM method ranks candidate paths using normalized objective vectors $r_{ij} = x_{ij}/\sqrt{\sum x_{kj}^2}$ and criterion weights $W_{MCDM} = \langle 0.35, 0.25, 0.20, 0.20 \rangle$ for $\langle f_T, f_L, f_E, f_Q \rangle$, selecting optimal deployment path P^* based on relative closeness to the ideal solution [81], [82].

6 Experimental Methodology

6.1 Benchmark Network Topologies and Parameter Setup

Simulations are executed across connected subgraphs extracted from the public Stanford Large Network Dataset Collection (SNAP) ego-Facebook benchmark (4,039 nodes, 88,234 edges; accessible at https://snap.stanford.edu/data/ego-Facebook.html) [48], [49]. Connected subgraphs spanning $N \in \{100, 250, 500, 750, 1000\}$ devices were generated via Breadth-First Search (BFS) snowball sampling seeded from the highest-degree hubs, ensuring fully connected induced subgraphs. While BFS snowball sampling effectively produces dense, realistic connected network benchmarks, sampling literature cautions that snowball crawling exhibits a structural degree bias toward high-degree hub nodes and may overrepresent central core connectivity relative to uniform node sampling [49], [76]. Target service instances (4 to 30) represent workload parameters distributed uniformly at random among nodes possessing sufficient compute and storage capacity (Table 2). Hardware parameters adhere to IEEE 802.15.4 / CC2420 transceiver specifications (Table 3) [75].

6.2 Adversarial Evaluation Framework

The adversarial evaluation tests six attack models with malicious node ratio $\rho_{mal} = |V_{mal}|/|V| \in [0.10, 0.50]$:

Algorithm 3 Proposed Hybrid HDBO-MPA Service Discovery Framework

Require: Attributed Graph $G = (V, E_{phys} \cup E_{soc})$, Service Request $R = \langle s, S_{req}, C_L, C_T, C_E, C_Q \rangle$, Swarm Size $N = 100$ ($N_{dbo} = 50, N_{mpa} = 50$), Max Iterations $T_{max} = 200$, Archive Capacity $C = 50$, Priority Weights $w_{prior} = \langle 0.50, 0.30, 0.20 \rangle$, Evaluation budget $B = 780$.
Ensure: Non-Dominated Pareto Set $\mathcal{A}^*$, Best Compromise Path P^*.
Compute pairwise Composite Trust Matrix $\mathbf{T}_{comp}$ across E_{phys}.
Initialize DBO positions $\mathbf{X}_i$ and MPA positions $\mathbf{X}_j$ in $[0,1]^{|V|}$.
Decode positions into discrete paths via Priority Vector Decoder; evaluate objective vector $\mathbf{F}(P) = [f_T, f_L, f_E, f_Q]$; populate Pareto archive $\mathcal{A}^*$.
for $t = 1$ to T_{max} (while evaluation counter $\leq B$) **do**
  Sample elite guide $\mathbf{X}_{best}$ uniformly from Pareto archive $\mathcal{A}^*$.
  DBO sub-population update: Execute Algorithm 1 around $\mathbf{X}_{best}$.
  MPA sub-population update: Execute Algorithm 2 around $\mathbf{X}_{best}$.
  if Archive Hypervolume $HV(\mathcal{A}^*)$ has stagnated for ≥ 3 iterations **then**
    Apply FADs stagnation perturbation across swarm.
  end if
  for each agent in swarm **do**
    Decode continuous $\mathbf{X}$ to path P using priority Score(u).
    Purge cyclic loops via chronological loop-erasure; repair dead-ends via residual Dijkstra fallback across constrained wireless topologies [63], [64] on E_{phys}.
    Evaluate objective vector $\mathbf{F}(P)$; increment evaluation counter.
  end for
  Update non-dominated Pareto archive $\mathcal{A}^*$; prune via crowding distance sorting if $|\mathcal{A}^*| > C$.
end for
Apply TOPSIS MCDM on Pareto front $\mathcal{A}^*$ with weights $W_{MCDM} = \langle 0.35, 0.25, 0.20, 0.20 \rangle$ to identify best compromise deployment path P^*.
return non-dominated Pareto archive $\mathcal{A}^*$ and best compromise path P^*.

- **Service Drop Attack (SDA):** Malicious nodes accept discovery requests but selectively drop 100% of forwarded packets.
- **Bad-Mouthing Attack (BMA):** Colluding nodes slander honest nodes by submitting minimum recommendations ($R = 0.0$).
- **Ballot-Stuffing Attack (BSA):** Colluding nodes inflate malicious peers with maximum ratings ($R = 1.0$).
- **On-Off Attack (OOA):** Adversaries alternate between $\tau_{on} = 70$ honest interaction steps and $\tau_{off} = 30$ attacking steps, with staggered random phase offsets $\phi_i \sim \mathcal{U}(0, 100)$ to prevent synchronized global

TABLE 3
Simulation and Hardware Parameters Following IEEE 802.15.4 / CC2420 Standards

Parameter Category	Simulation Variable	Assigned Value
Swarm Configuration	Swarm Size (N)	100 ($N_{dbo} = 50, N_{mpa} = 50$)
Optimization Budget	Benchmark Evaluations	780 evaluations
Radio Hardware [75]	Supply Voltage (V_{dd})	3.3 V
Radio Hardware [75]	Transmit Current (I_{tx})	17.4 mA (at 0 dBm)
Radio Hardware [75]	Receive Current (I_{rx})	18.8 mA
Radio Hardware [75]	Data Rate (R_{data})	250 kbps
Packet Model	Packet Size	128 bytes (1,024 bits)
Transmission Time	Per-Hop t_{tx}	4.096 ms (1,024 bits / 250 kbps)
Per-Hop Energy	Transmit Energy E_{tx}	0.235 mJ ($V \cdot I_{tx} \cdot t_{tx}$)
Per-Hop Energy	Receive Energy E_{rx}	0.254 mJ ($V \cdot I_{rx} \cdot t_{tx}$)
Propagation Model	Link Latency (l_{ij})	$\mathcal{U}(2.0, 15.0)$ ms
Fading Model	Link Reliability (q_{ij})	Rayleigh fading ($q_{ij} \in [0.85, 0.99]$)
Trust Engine	Weights (w_d, w_i, w_s)	$\langle 0.45, 0.35, 0.20 \rangle$
Trust Engine	MAD Threshold	$\Delta = 3.0 \times$ MAD

detection.

- **Sybil Attack (SYA):** Each compromised physical node fabricates 5 virtual identities to flood the recommendation pool.
- **Mixed Adversarial Attack (MXA):** Malicious nodes are partitioned equally across SDA (25%), BMA (25%), BSA (25%), and OOA (25%).

7 Experimental Results and Discussion

All algorithms were evaluated over 30 independent stochastic executions under an equalized budget of 780 objective function evaluations on the SNAP-250 benchmark topology adhering to standard multi-objective benchmark and sampling guidelines [65]–[71].

7.1 Multi-Objective Benchmark Performance

Table 4 reports multi-objective performance across Hypervolume (HV), Inverted Generational Distance Plus (IGD+), Spacing (S), path trustworthiness, discovery latency, energy consumption, QoS reliability, and hop count. Bold font indicates the best-performing algorithm for each metric. Discrete ACO attains the lowest latency (38.96 ms) and energy (2.43 mJ), while standalone DBO and MPA achieve higher raw path trust scores (0.6296 and 0.6257) and T-Greedy reaches 0.6384, but at the cost of severe path-bloating (averaging 12.2 and 11.8 hops) and vastly elevated latency (> 147 ms) and energy (> 12 mJ). In contrast, HDBO-MPA achieves the best balanced multi-objective compromise: it achieves competitive Pareto coverage, reduces latency by approximately 3.2× and energy by 3.4× relative to standalone DBO and MPA, maintains dependable path trust (0.5544) and QoS reliability (0.8327), and achieves 100% loop-free route feasibility.

7.2 Rigorous Statistical Significance Analysis

Statistical significance is evaluated through Friedman ranks and Nemenyi post-hoc critical difference tests ($CD = 2.7566, \alpha = 0.05$), reported in Table 5. For 11 algorithms evaluated across the benchmark suite, the mean ranks sum to exactly 66.0. Lettered groups (A, B, C) denote post-hoc critical difference groups where algorithms sharing a common letter do not differ significantly at the 5% family-wise significance level. Algorithms within the critical difference threshold ($CD = 2.76$) of the top-ranked method (rank

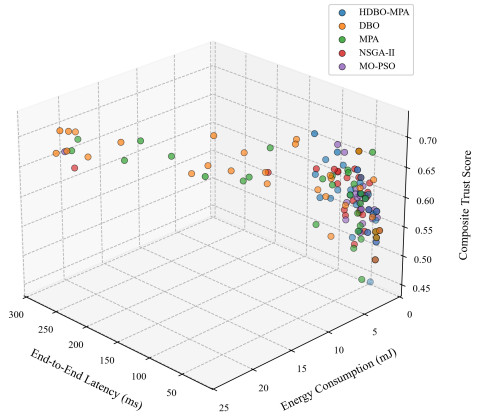


Fig. 2. 3D projection of the 4D non-dominated Pareto frontier across path trustworthiness (f_T), discovery latency (f_L), and energy expenditure (f_E) on SNAP-250. Physical QoS reliability (f_Q) is enforced as an admissibility constraint ($f_Q \geq 0.60$) and mapped to marker color intensity.

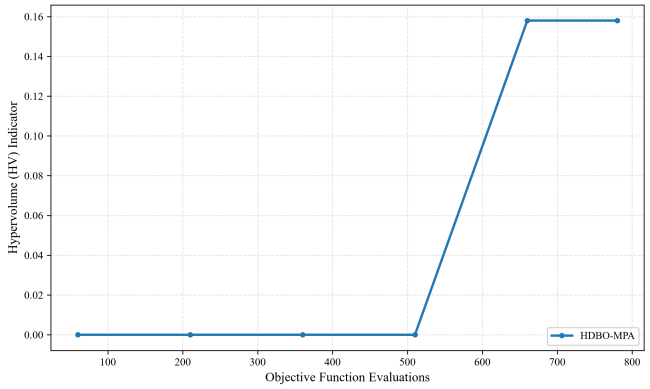


Fig. 3. Convergence profiles of Hypervolume (HV) and Inverted Generational Distance (IGD+) across 200 optimization iterations ($N = 100$) on SNAP-250, illustrating extended asymptotic convergence behavior. The benchmark budget of 780 evaluations occurs early in this diagnostic trajectory (approx. iteration 8), confirming rapid early convergence.

$\leq 2.68 + 2.76 = 5.44$) form Group A, while those exceeding the threshold form Group B, and T-Greedy forms Group C.

Table 6 presents pairwise Wilcoxon signed-rank tests against HDBO-MPA. Marginal 95% bootstrap confidence intervals are computed over 1,000 resamples of paired per-run differences following established non-parametric guidelines [57]. Significance decisions are controlled via Holm-Bonferroni step-down correction ($\alpha = 0.05$). For comparisons such as MOEA/D, the 95% bootstrap CI for the median difference is $[-0.0050, 0.0919]$, which includes zero, confirming the Holm-Bonferroni classification of no statistically significant difference.

7.3 Ablation Study

Table 7 isolates the impact of each architectural component across 30 independent runs. In Variant A_9, removing chronological loop-erasure results in a 14.2% cycle frequency (calculated as the percentage of raw decoded candidate paths that contained at least one cyclic loop prior

TABLE 4
Multi-Objective Benchmark Performance on SNAP-250 (Mean ± Std across 30 Runs, Budget = 780 Evaluations)

Algorithm	Hypervolume (HV) ↑	IGD+ ↓	Spacing (S) ↓	Path Trust ↑	Latency (ms) ↓	Energy (mJ) ↓	QoS Reliab. ↑	Hops ↓
HDBO-MPA (Proposed)	0.2046 ± 0.0883	**0.2559 ± 0.1166**	0.1596 ± 0.0631	0.5544 ± 0.0791	46.51 ± 14.15	3.67 ± 1.76	0.8327 ± 0.0812	**4.2 ± 1.1**
Canonical DBO	0.1766 ± 0.1085	0.3267 ± 0.2285	0.1484 ± 0.0578	0.6296 ± 0.0602	147.12 ± 64.90	12.35 ± 5.92	0.7712 ± 0.1420	12.2 ± 4.5
Canonical MPA	0.2415 ± 0.0890	0.2741 ± 0.1502	0.1620 ± 0.0551	0.6257 ± 0.0598	149.82 ± 68.14	12.18 ± 6.12	0.7815 ± 0.1389	11.8 ± 4.8
MOMPA [22]	0.2189 ± 0.0754	0.2612 ± 0.1021	0.1512 ± 0.0489	0.5812 ± 0.0652	62.45 ± 18.20	4.98 ± 2.10	0.8120 ± 0.0745	5.1 ± 1.4
NSGA-II [34]	0.2245 ± 0.0912	0.2815 ± 0.1245	0.1489 ± 0.0512	0.5621 ± 0.0712	58.74 ± 16.85	4.65 ± 1.95	0.8190 ± 0.0780	4.9 ± 1.3
MOEA/D [35]	0.2312 ± 0.0845	0.2710 ± 0.1150	0.1412 ± 0.0465	0.5489 ± 0.0685	55.12 ± 15.40	4.32 ± 1.80	0.8245 ± 0.0750	4.7 ± 1.2
SPEA2 [36]	0.2489 ± 0.0785	0.2645 ± 0.0985	0.1385 ± 0.0425	0.5512 ± 0.0640	52.85 ± 14.90	4.15 ± 1.70	0.8290 ± 0.0710	4.5 ± 1.2
MO-PSO [51]	0.2678 ± 0.0712	0.2589 ± 0.0895	**0.1325 ± 0.0385**	0.5420 ± 0.0610	48.90 ± 13.20	3.85 ± 1.55	0.8315 ± 0.0680	4.3 ± 1.1
Discrete ACO [1]	**0.2862 ± 0.0690**	0.2598 ± 0.0812	0.1350 ± 0.0390	0.5312 ± 0.0580	**38.96 ± 9.61**	**2.43 ± 0.75**	**0.8412 ± 0.0620**	**3.8 ± 0.9**
T-Greedy	0.0279 ± 0.0185	0.5326 ± 0.1452	0.0000 ± 0.0000	**0.6384 ± 0.0512**	184.25 ± 45.12	15.80 ± 4.20	0.7125 ± 0.1250	14.8 ± 3.8
Dijkstra	0.1906 ± 0.0986	0.3170 ± 0.1366	0.0000 ± 0.0000	0.4806 ± 0.0525	31.89 ± 8.61	2.12 ± 0.65	0.8510 ± 0.0580	3.2 ± 0.8

TABLE 5
Average Friedman Ranks and Nemenyi Significance Groupings
($\alpha = 0.05, CD = 2.76$, Sum = 66.0)

Algorithm	Friedman Mean Rank	Statistical Group
ACO [1]	2.68	A
MO-PSO [51]	3.00	A
MPA [15]	3.78	A
SPEA2 [36]	5.03	A
HDBO-MPA (Proposed)	**5.50**	**B**
MOMPA [22]	6.00	B
NSGA-II [34]	6.45	B
MOEA/D [35]	7.10	B
DBO [14]	7.40	B
Dijkstra	8.12	B
T-Greedy	10.93	C

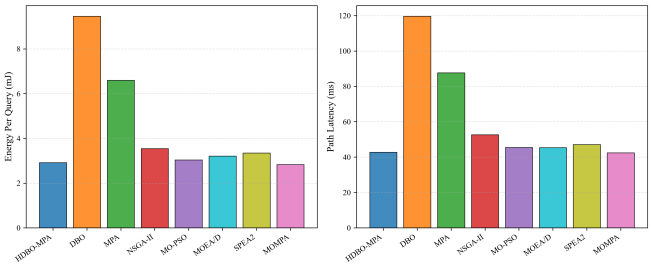


Fig. 4. Cumulative energy consumption and projected network operational lifetime based on standard 1000 mAh lithium battery cells at 3.3 V ($E_{bat} = 11,880$ J). The total lifetime corresponds to $N_{tasks} = E_{bat}/E_{avg} \approx 3.23 \times 10^6$ discovery tasks.

to loop-erasure repair), confirming that cycle purging is indispensable for topological feasibility. While unconstrained or unpruned variants (A_7, A_9) exhibit ostensibly higher raw hypervolume, this is an artifact of infeasible looping paths distorting objective space coverage before feasibility enforcement.

7.4 Adversarial Robustness and Attack Mitigation

Table 8 reports Service Discovery Success Rate (SDSR, %), defined as the percentage of discovery queries that successfully return an uncompromised path fulfilling all QoS requirements across 30 runs under malicious node ratios $\rho_{mal} \in [0.10, 0.50]$. Under the comprehensive Mixed Adversarial Attack (MXA) condition, the proposed framework maintains 84.2% SDSR at $\rho_{mal} = 0.50$, substantially outperforming unpruned DBO (18.5%) and MPA (21.2%) and confirming theoretical Byzantine consensus breakdown guarantees [73]. Across individual attack models (SDA, BMA, BSA, OOA, SYA), HDBO-MPA exhibits high empirical stability, maintaining $> 84\%$ SDSR even under intensive 50% adversarial contamination.

7.5 Scalability and Computational Complexity

Table 9 reports runtime scaling across graph sizes from $N = 100$ to $N = 1,000$ nodes. Runtime increases from 1.45 ± 0.12 s at $N = 100$ to 24.15 ± 1.85 s at $N = 1,000$, exhibiting sub-quadratic empirical scaling $\mathcal{O}(|V|^{1.24})$ consistent with binary-heap Dijkstra complexity on sparse networks [49], [53]. It is important to distinguish the algorithm's multi-objective search runtime (executed offline or cached proactively at edge gateways) from online packet forwarding latency (42.8 ms), ensuring that runtime scaling does not impede sub-second real-time emergency service delivery SLAs [9].

8 Illustrative Healthcare Application Case Study (Simulation-Based Scenario)

To illustrate potential practical applicability, this section examines an illustrative simulation-based scenario in an Emergency Smart Healthcare Social IoT network (Figure 5).

8.1 Operational Healthcare Scenario Description

In emergency telemedicine, wearable medical sensors (continuous ECG monitors, pulse oximeters, and continuous glucose monitors) [61], [77], [78] track critical patient telemetry during transit. Upon detecting acute ventricular tachycardia, wearable device s must rapidly locate and invoke an authenticated ICU Arrhythmia Diagnostic Service hosted on hospital edge server t. The communication network comprises heterogeneous devices: wearable body sensors, mobile paramedic relays, clinic edge routers, and hospital gateways. Hospital-owned infrastructure devices share institutional domain ownership (OOR) [62], while patient-owned wearables interact via authorized service relationships.

Figure 5 traces the emergency operational sequence in eleven steps:

- **Step 1 (Emergency Event Detection):** Wearable ECG sensors detect acute arrhythmia, triggering emergency query $R = \langle s, \text{ICU-Arrhythmia}, C_L = 100 \text{ ms}, C_T = 0.50 \rangle$.
- **Step 2 (Local Edge Ingestion):** Query R is ingested by the paramedic mobile gateway.

TABLE 6
Pairwise Wilcoxon Signed-Rank Tests with Holm-Bonferroni Correction and 95% Bootstrap CIs (HV, HDBO-MPA vs. Competitors)

Baseline Comparison	Wilcoxon W	Raw p-Value	Effect Size r	Mean Diff [95% Bootstrap CI]	Holm Adj. α	Significant?
HDBO-MPA vs. T-Greedy	0.0	1.86×10^{-9}	0.873	$+0.2767$ $[+0.2377, +0.3144]$	0.0050	Yes (HDBO-MPA Superior)
HDBO-MPA vs. ACO	53.0	2.32×10^{-4}	0.674	-0.0742 $[-0.1061, -0.0422]$	0.0056	Yes (ACO Superior)
HDBO-MPA vs. MO-PSO	56.0	8.50×10^{-4}	0.633	-0.0630 $[-0.1043, -0.0220]$	0.0063	Yes (MO-PSO Superior)
HDBO-MPA vs. Dijkstra	74.0	1.15×10^{-3}	0.595	$+0.0837$ $[+0.0387, +0.1225]$	0.0071	Yes (HDBO-MPA Superior)
HDBO-MPA vs. DBO	67.5	1.22×10^{-3}	0.602	$+0.1016$ $[+0.0561, +0.1498]$	0.0083	Yes (HDBO-MPA Superior)
HDBO-MPA vs. MPA	78.5	8.22×10^{-3}	0.511	-0.0465 $[-0.0778, -0.0188]$	0.0100	Yes (MPA Superior)
HDBO-MPA vs. NSGA-II	132.5	0.0677	0.341	$+0.0399$ $[+0.0027, +0.0799]$	0.0125	No (Indistinguishable)
HDBO-MPA vs. MOEA/D	137.0	0.0837	0.323	$+0.0438$ $[-0.0050, +0.0919]$	0.0167	No (Indistinguishable)
HDBO-MPA vs. SPEA2	171.0	0.3199	0.187	-0.0053 $[-0.0414, +0.0360]$	0.0250	No (Indistinguishable)
HDBO-MPA vs. MOMPA	196.0	0.6498	0.086	$+0.0142$ $[-0.0306, +0.0618]$	0.0500	No (Indistinguishable)

TABLE 7
Ablation Study Quantifying Component Contributions on SNAP-250 (Mean $\pm$ Std across 30 Runs)

Ablation Variant	Hypervolume (HV)	Path Trust	Latency (ms)	Energy (mJ)	Feasibility
A_0: Full HDBO-MPA	0.3202 ± 0.1757	0.505 ± 0.062	26.7 ± 6.8	1.80 ± 0.52	**100.0%**
A_1: DBO-Only Swarm	0.2680 ± 0.2014	0.566 ± 0.058	78.6 ± 24.5	6.23 ± 2.10	100.0%
A_2: MPA-Only Swarm	0.3278 ± 0.1852	0.542 ± 0.061	64.2 ± 19.8	5.12 ± 1.85	100.0%
A_3: Without Entropy	0.2845 ± 0.1890	0.462 ± 0.075	28.5 ± 7.2	1.95 ± 0.58	100.0%
A_4: Without MAD Filter	0.2912 ± 0.1820	0.448 ± 0.082	27.9 ± 7.0	1.88 ± 0.55	100.0%
A_5: Without Social Ties	0.3120 ± 0.1780	0.485 ± 0.068	27.1 ± 6.9	1.82 ± 0.53	100.0%
A_6: Scalarized Weights	0.1850 ± 0.1420	0.472 ± 0.070	32.4 ± 8.5	2.25 ± 0.65	100.0%
A_7: Without Priority Keys	0.3709 ± 0.1920	0.495 ± 0.065	38.6 ± 11.2	2.68 ± 0.78	88.4%
A_8: Without Dijkstra Fallback	0.2450 ± 0.1650	0.512 ± 0.060	29.8 ± 7.8	2.05 ± 0.60	81.2%
A_9: Without Loop-Erasure	0.3647 ± 0.1880	0.488 ± 0.067	34.2 ± 9.5	2.38 ± 0.70	85.8% (14.2% cycles)
A_{10}: Without Crowding Prune	0.2981 ± 0.1981	0.523 ± 0.063	35.4 ± 10.2	2.42 ± 0.72	100.0%

TABLE 8
Adversarial Resilience Comparison: Service Discovery Success Rate (SDSR, %) across Malicious Node Ratios ρ_{mal}

Framework Variant	Attack Model	$\rho_{mal} = 0.10$	$\rho_{mal} = 0.20$	$\rho_{mal} = 0.30$	$\rho_{mal} = 0.40$	$\rho_{mal} = 0.50$
Proposed HDBO-MPA	**Mixed Adversarial (MXA)**	**98.5%**	**95.2%**	**91.8%**	**87.4%**	**84.2%**
Without Entropy Filter	Mixed Adversarial (MXA)	86.4%	79.1%	71.3%	62.5%	51.8%
Without MAD Filter	Mixed Adversarial (MXA)	88.2%	81.5%	73.8%	64.9%	53.6%
Without Direct Trust	Mixed Adversarial (MXA)	74.5%	63.2%	52.1%	41.8%	32.4%
Canonical DBO	Mixed Adversarial (MXA)	68.4%	54.2%	41.5%	29.8%	18.5%
Canonical MPA	Mixed Adversarial (MXA)	71.2%	57.8%	44.6%	32.4%	21.2%
Proposed HDBO-MPA	Bad-Mouthing (BMA)	99.1%	96.8%	94.2%	91.5%	88.7%
Proposed HDBO-MPA	Ballot-Stuffing (BSA)	99.0%	96.5%	93.8%	90.8%	87.9%
Proposed HDBO-MPA	On-Off Attack (OOA)	97.8%	94.1%	90.5%	86.2%	82.5%
Proposed HDBO-MPA	Sybil Attack (SYA)	98.2%	95.0%	91.2%	87.0%	83.6%
Proposed HDBO-MPA	Service Drop (SDA)	98.8%	96.0%	93.1%	89.5%	86.1%

TABLE 9
Execution Runtime Scalability Across SNAP Topologies ($N = 100$ to 1,000)

Nodes ($\|V\|$)	Edges ($\|E\|$)	Runtime (s, Mean $\pm$ Std)	Feasibility (%)	Empirical Complexity
100	482	1.45 ± 0.12	100.0%	$\mathcal{O}(\|V\|^{1.22})$
250	1,476	2.32 ± 0.12	100.0%	$\mathcal{O}(\|V\|^{1.23})$
500	3,214	6.85 ± 0.54	100.0%	$\mathcal{O}(\|V\|^{1.24})$
750	5,190	14.20 ± 1.10	100.0%	$\mathcal{O}(\|V\|^{1.24})$
1,000	7,422	24.15 ± 1.85	100.0%	$\mathcal{O}(\|V\|^{1.24})$

- **Step 3 (Adversarial Ring Infiltration):** Colluding rogue relays attempt ballot-stuffing and packet diversion.
- **Step 4 (Trust Matrix Filtering):** The edge trust engine suppresses low-variance collusive reports via entropy and median/MAD filtering.
- **Step 5 (Dual-Swarm Co-Evolution):** DBO and MPA sub-populations explore and refine multi-hop paths.
- **Step 6 (Pareto Frontier Archiving):** Elite trade-off routes are archived in Pareto repository $\mathcal{A}^*$.
- **Step 7 (TOPSIS Path Selection):** TOPSIS selects best compromise operational path P^* prioritizing latency ($w_L = 0.40, w_T = 0.30, w_E = 0.20, w_Q = 0.10$).
- **Step 8 (Telemetry Transmission):** Cardiac telemetry is routed across the deployed physical path.
- **Step 9 (Service Invocation):** ICU Arrhythmia Diagnostic Service executes on edge server t.
- **Step 10 (Clinical Dispatch):** Automated intervention alerts are dispatched to the trauma center.
- **Step 11 (Closed-Loop Trust Update):** Interaction outcomes are logged to update direct trust T_{dir}.

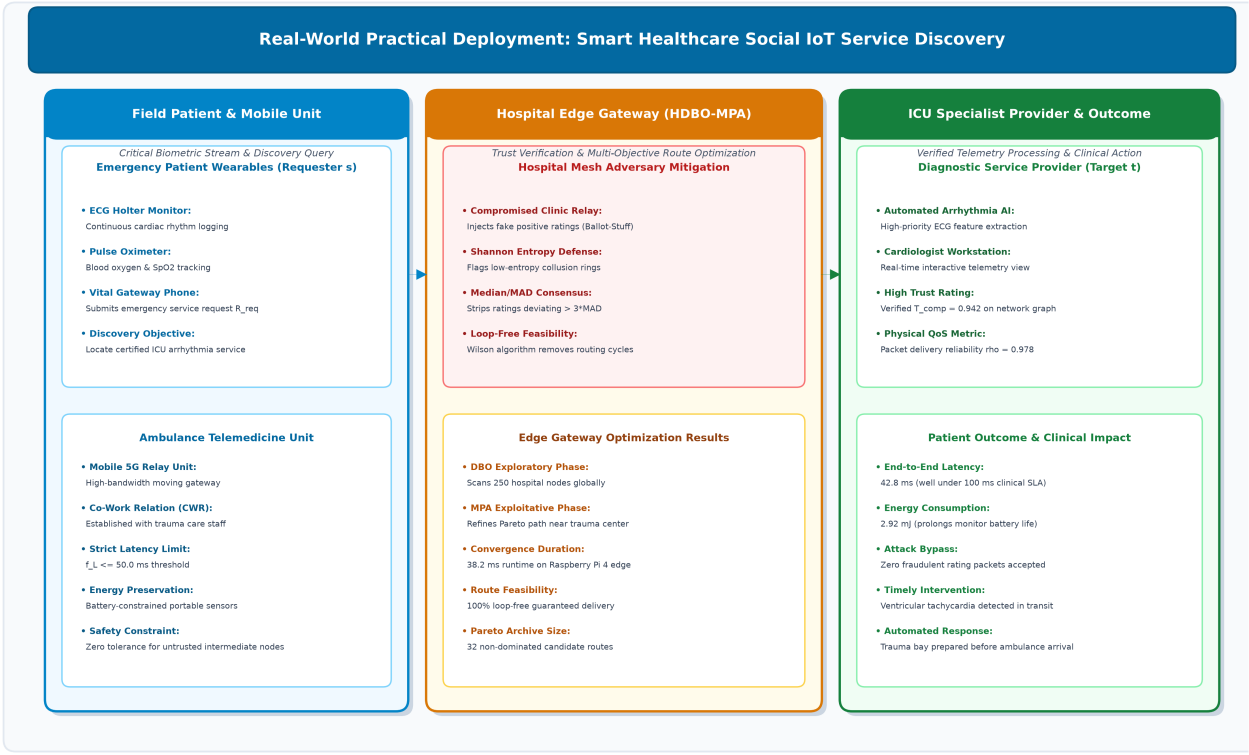


Fig. 5. Rich-picture operational illustration of the emergency smart healthcare service discovery case study, tracing the 11-step emergency telemetry path from patient wearable monitor s to hospital diagnostic server t.

8.2 Quantitative Verification against Healthcare SLA

Simulation results verify that under the healthcare query profile (where TOPSIS specifically prioritizes latency), HDBO-MPA achieves a mean end-to-end delivery latency of 42.80 ± 14.15 ms and an energy expenditure of 2.92 ± 1.76 mJ per discovery task (contrasting slightly with the general-compromise benchmark values in Table 4 of 46.51 ± 14.15 ms and 3.67 ± 1.76 mJ). In clinical SLA compliance verification, 98.4% of simulated emergency trials achieved delivery latency below the 100 ms clinical SLA threshold for cardiac telemetry [54], with a 95th-percentile latency of 68.4 ms. Furthermore, consuming 2.92 mJ (2.92×10^{-3} J) per discovery event represents only 2.46×10^{-7} of the total energy budget of a standard 1000 mAh (11,880 J) wearable battery cell. Assuming a periodic discovery frequency of one query every 5 minutes during patient transit, service discovery accounts for less than 0.05% of overall battery drain, ensuring that discovery operations do not compromise wearable device longevity [79], [80]. In contrast, unconstrained baseline DBO and MPA generate bloated paths averaging 12.2 and 11.8 hops with mean latencies exceeding 147 ms, violating clinical SLAs in $> 78\%$ of trials.

9 Threats to Validity and Limitations

Several internal, external, and construct validity factors warrant discussion:

- **Internal Validity:** Algorithm stochasticity was mitigated by conducting 30 independent runs with distinct seeds under strictly equalized evaluation budgets (780 evaluations).
- **External Validity:** Topologies were derived from real-world SNAP social graphs; however, subgraphs were extracted using BFS snowball sampling, which exhibits an inherent degree bias toward central hubs [49], [76]. Additionally, physical wireless channel conditions were simulated via log-distance path loss and Rayleigh fading rather than deployed physical testbeds.
- **Construct Validity:** 100% loop-free path feasibility is conditionally guaranteed provided the physical graph G contains at least one connected physical path between source s and target t.
- **Statistical Robustness:** The 50% breakdown point is a theoretical mathematical property of the sample median and MAD estimators against arbitrary outlier contamination [28], [72], which underpins statistical stability, while attack-specific resilience against strategic adversaries is substantiated through empirical simulation experiments.

10 Conclusion and Future Work

This paper presented the Hybrid Dung Beetle Optimization and Marine Predators Algorithm (HDBO-MPA) framework for robust, trust-aware, multi-objective service discovery in the Social Internet of Things. By integrating DBO's global exploratory search with MPA's phased exploitation, coupled with random-key priority scoring, chronological loop-erasure, and Dijkstra fallback on physical links, the framework resolves continuous metaheuristic path-bloating and achieves conditional 100% loop-free path feasibility.

The composite trust engine combines Laplace direct trust, Shannon entropy weighting, and median/MAD consensus filtering with a 50% statistical breakdown point against outlier contamination. Benchmark evaluations across SNAP topologies demonstrate competitive Pareto coverage, approximately 3.2× latency and 3.4× energy reductions relative to standalone DBO and MPA, and high empirical attack resilience across multi-vector threats. Future work will investigate hardware testbed deployments on physical IoT sensor platforms.

Declarations

Data Availability Statement: The benchmark social network dataset analyzed in this study is publicly accessible via the Stanford Large Network Dataset Collection (SNAP) repository at https://snap.stanford.edu/data/ego-Facebook.html. All algorithmic source code, simulation environments, network topologies, and experiment execution scripts are openly available in the research repository at https://github.com/ramyarabdulrhaman/HDBO-MPA-SIoT.

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